

A MODEL FOR EXAMINING LOCAL GEOGRAPHIC VARIATION IN OUT-OF-HOSPITAL CARDIAC ARREST OUTCOMES

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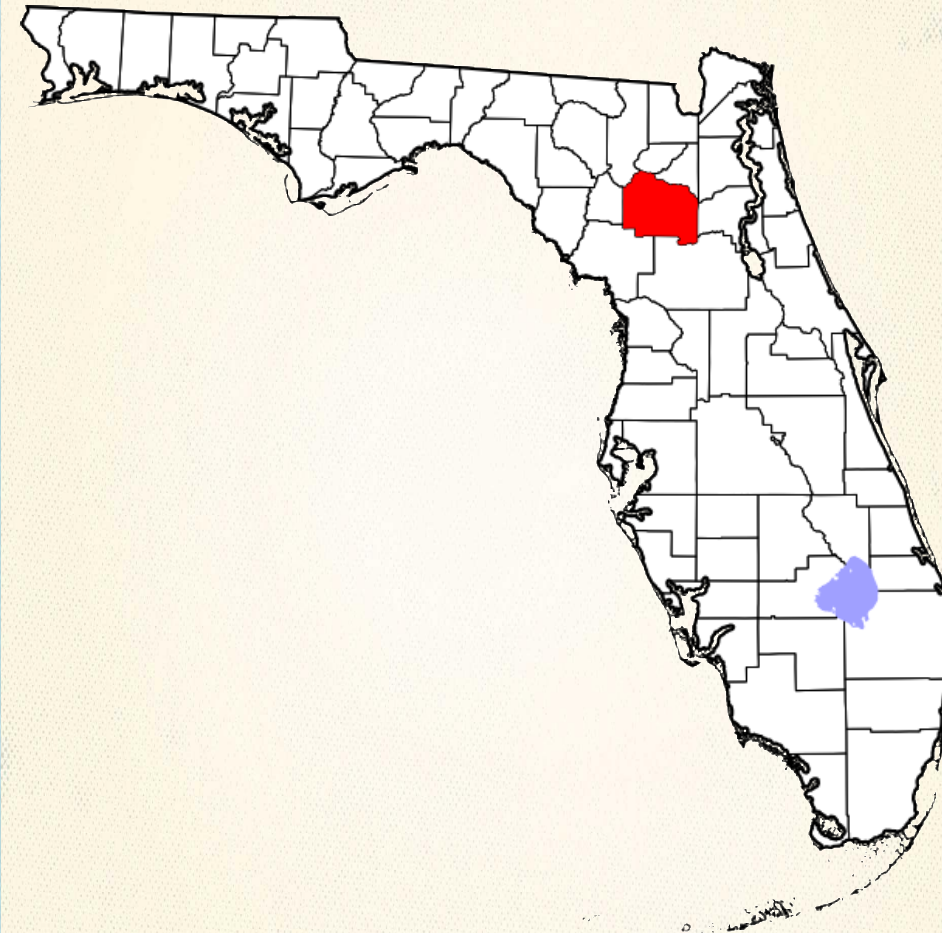
Background

- Overall rates of survival (<10%) have remained consistent in the last 3 decades
- US regional survival rates have been reported to range from 3% to 40%
- “Much work remains to understand regional differences” (AHA, 2018)

Objectives

- Assess sub-county geographic patterns in OHCA outcomes at key survival milestones
- Identify patient- and neighborhood-level factors that influence these outcomes
- Build a replicable model to allow communities to identify local patterns (Precision Public Health)

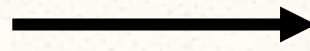
*Growing demand for
community-oriented
assessment strategies*



Methods

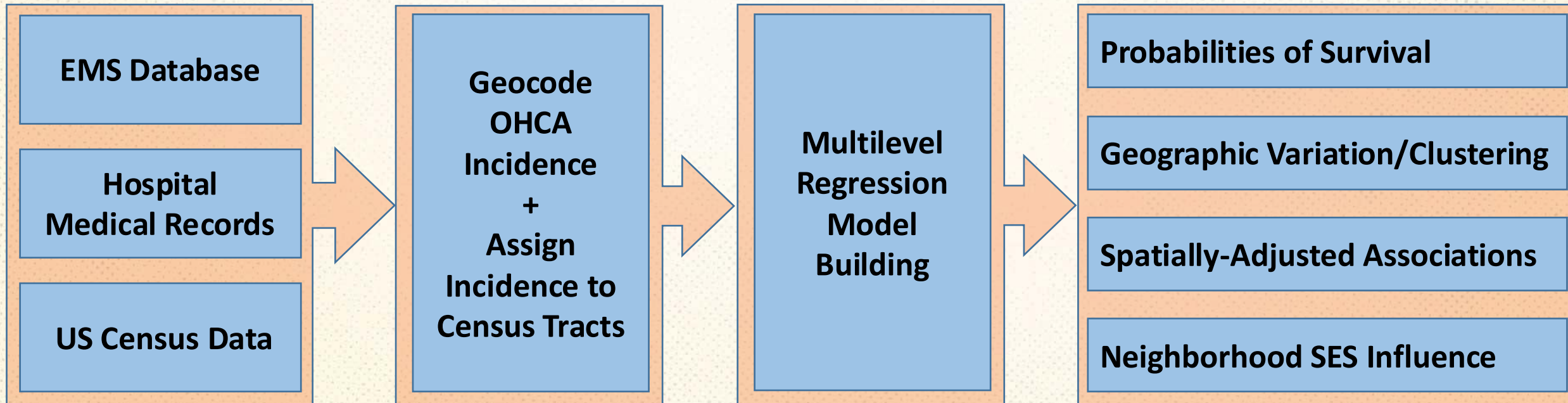
- Collaboration between UF Health ED and Alachua County Fire Rescue
- **Inclusion:** Adult, Non-Traumatic OHCA from 2012-2016
- **Survival Outcomes:** (1) ED Admit (2) Hospital Admit (3) Discharge (4) Home Discharge

Prehospital Sample
(n=1335)



In-hospital Sample
(n=372)

Model Overview



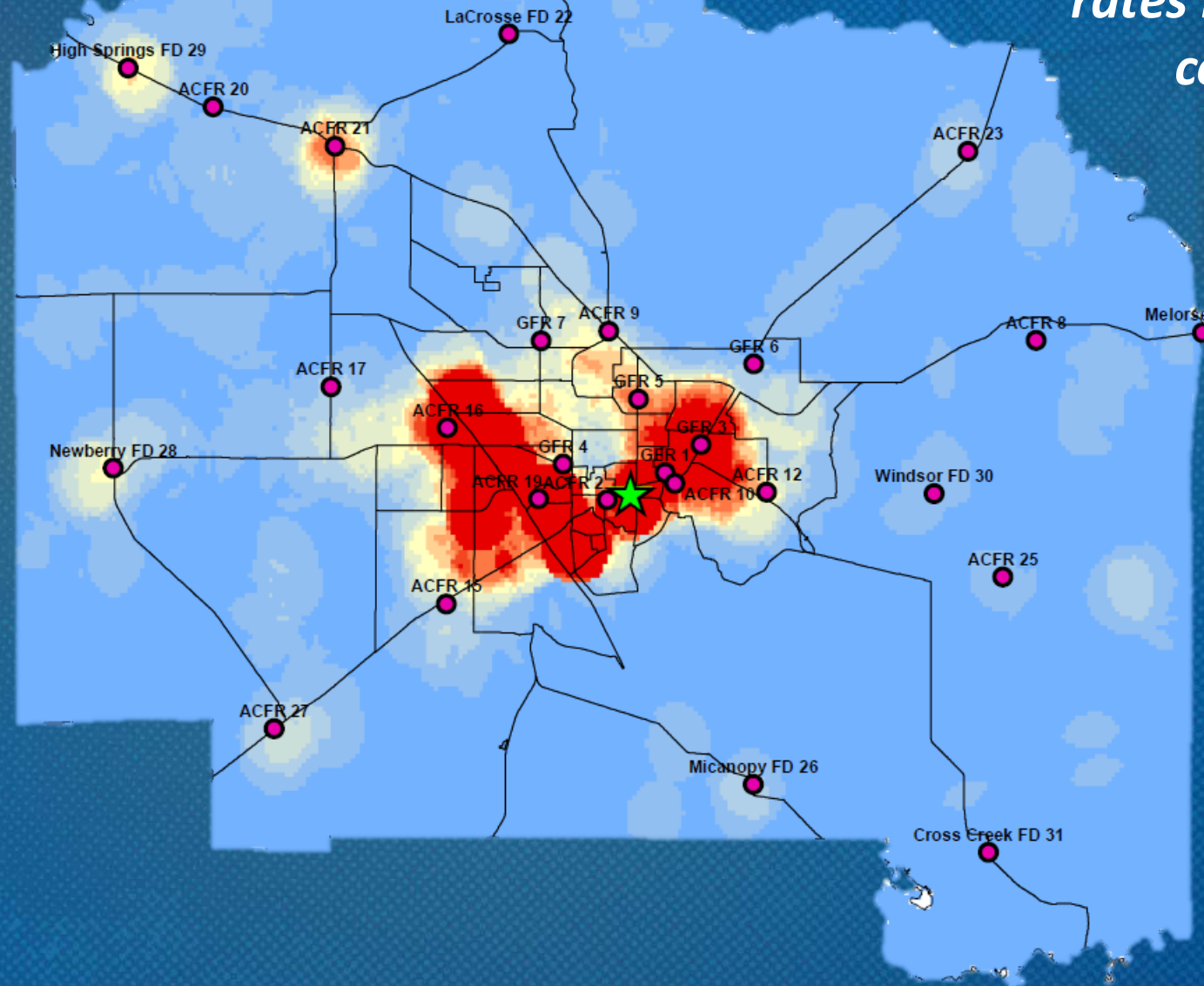
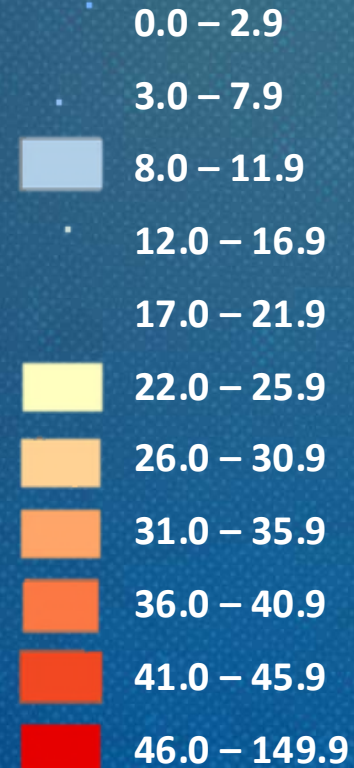
HEAT MAP OF OHCA INCIDENCE (n=1335)

UF

*5-Year OHCA incidence
rates is significantly
correlated with
geography*

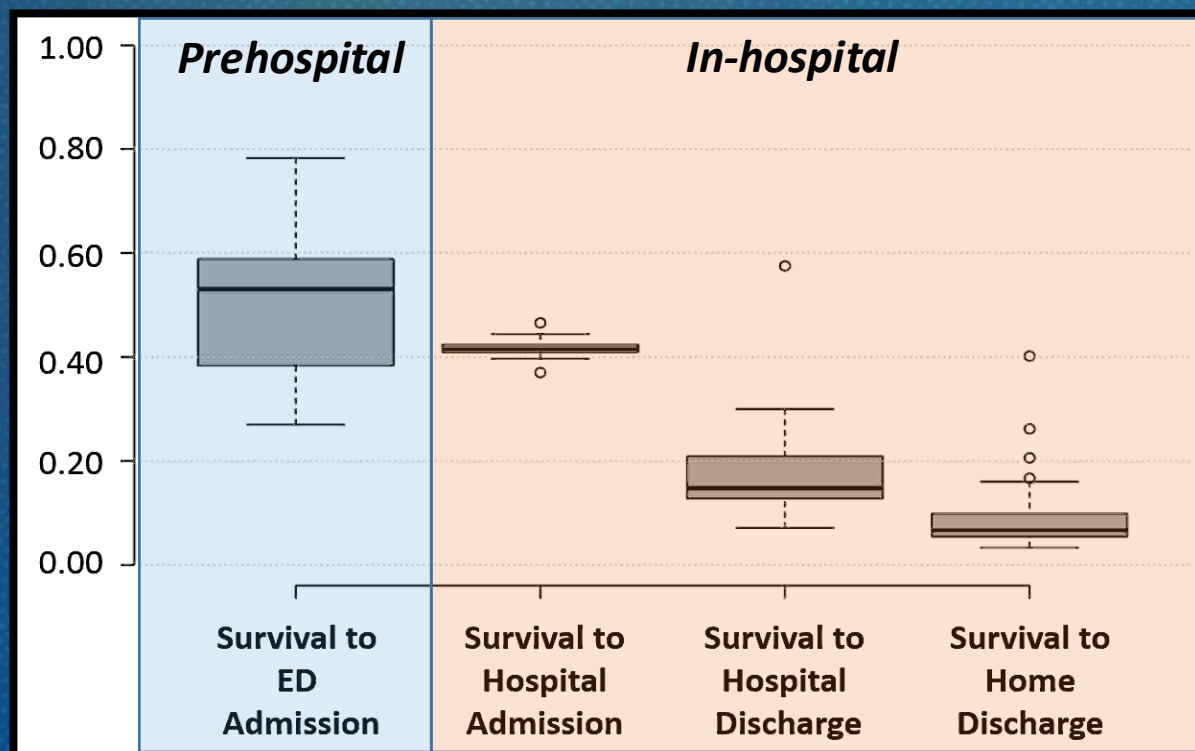
5-year Incidence
per sq. km.

Legend



RESULTS

Outcome	Probability (95%CI)
Survival to ED Admission	57.0% (51.3% - 62.6%)
Survival to Hospital Admission	41.7% (36.1% - 47.6%)
Survival to Discharge	16.1% (10.7% - 23.5%)
Survival to Home Discharge	7.1% (3.7% - 13.3%)



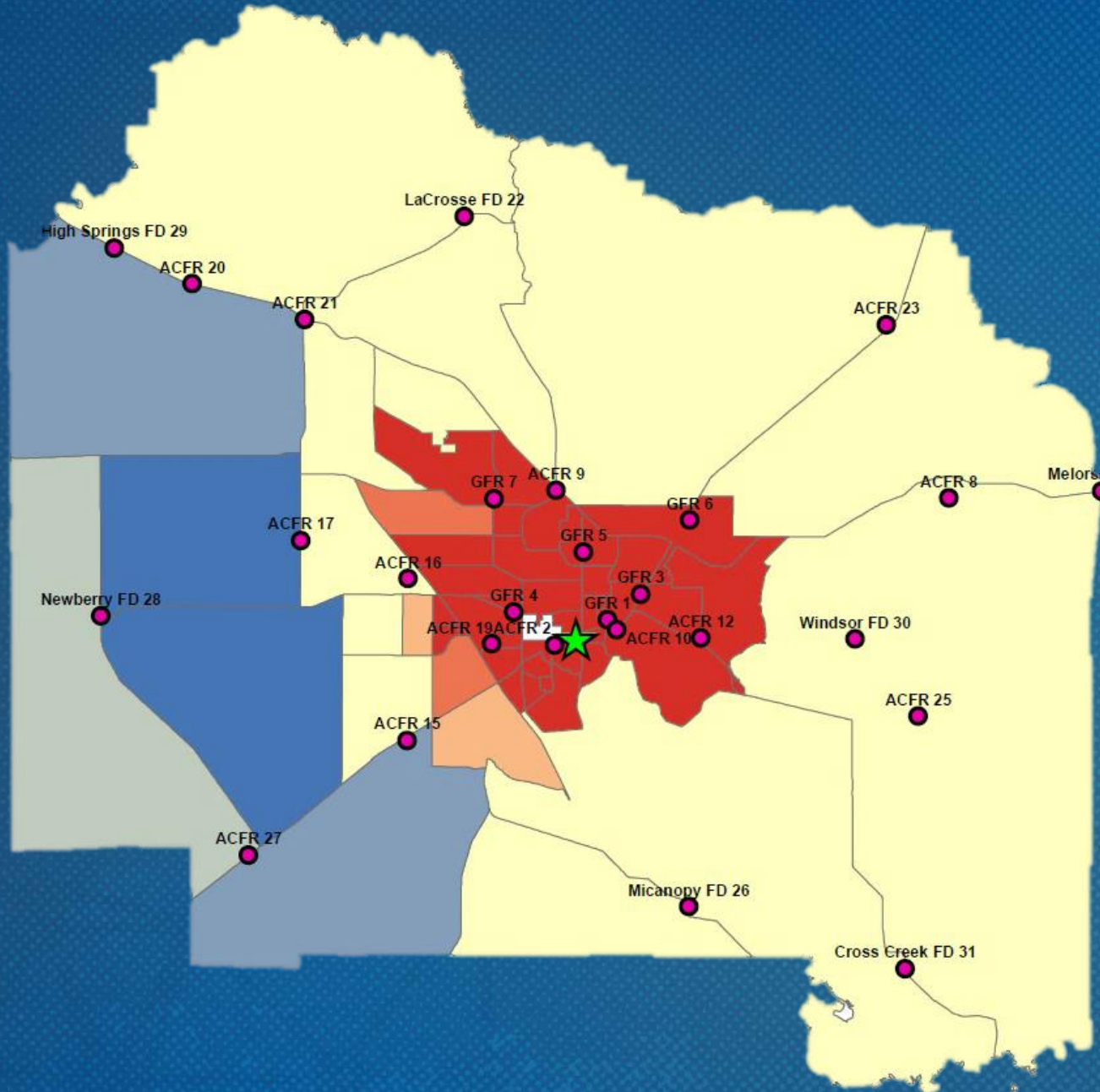
- *Significant geographic variation in survival to ED, discharge, and discharge home rates*
- *Significant geographic clustering in survival to ED rates*

SURVIVAL TO ED HOT SPOT ANALYSIS

Getis – Ord Gi*

Legend

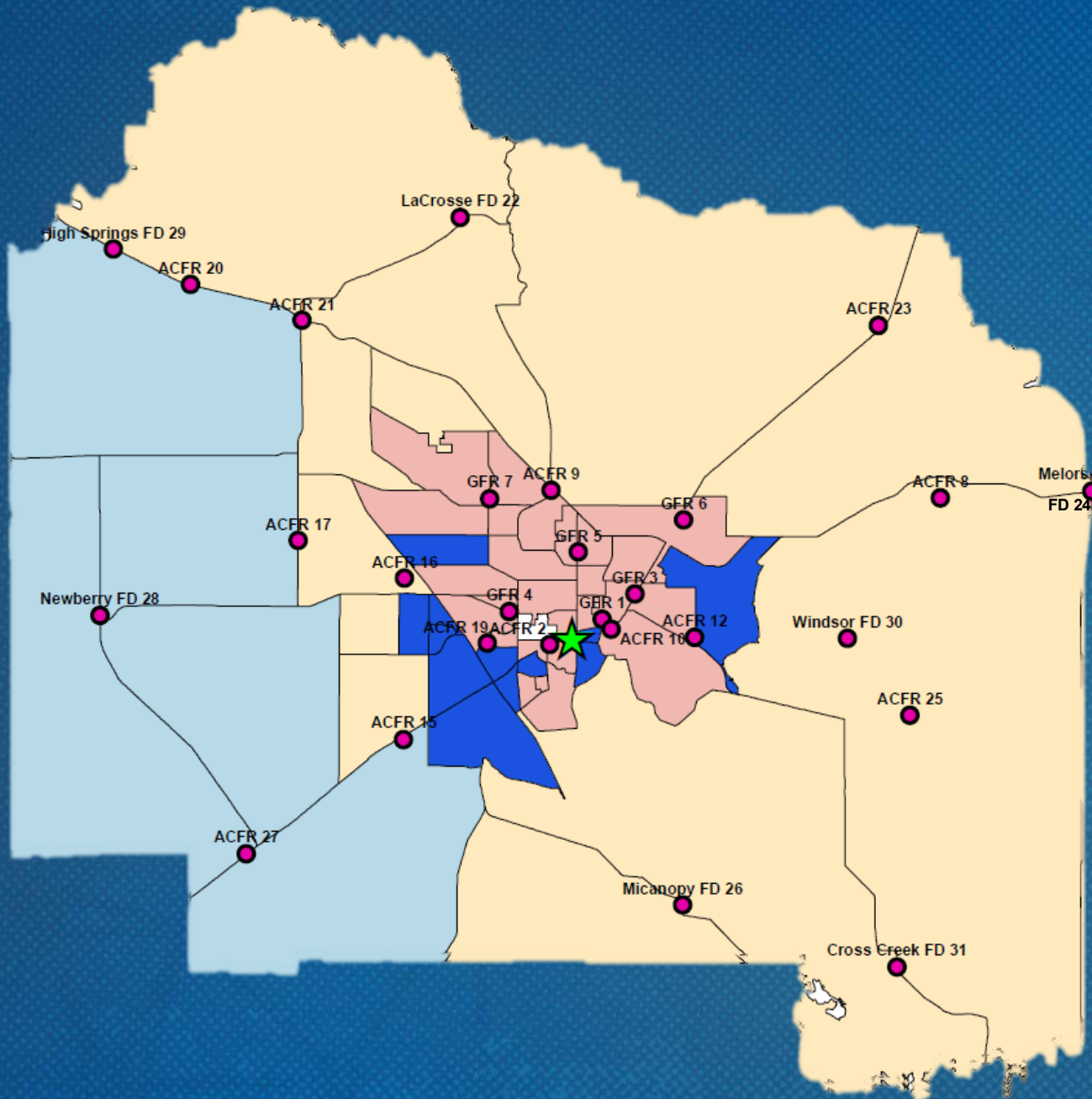
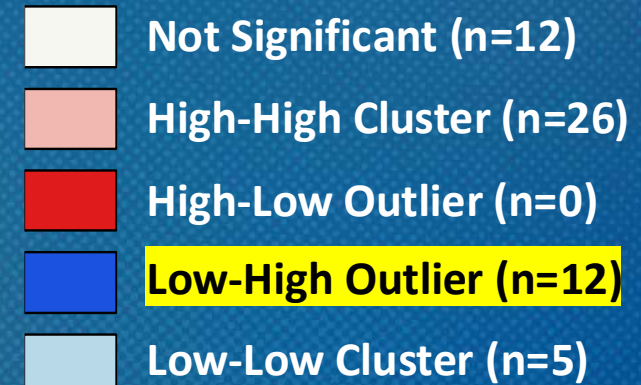
-  Cold Spot – 99% Confidence (2)
-  Cold Spot – 95% Confidence (2)
-  Cold Spot – 90% Confidence (1)
-  Not Significant (12)
-  Hot Spot – 90% Confidence (2)
-  Hot Spot – 95% Confidence (2)
-  Hot Spot – 99% Confidence (34)



SURVIVAL TO ED OUTLIER ANALYSIS

Local Moran's I

Legend



How can we explain this variation?

Patient Level Predictors	Neighborhood Level Predictors
Age	Percent Living Alone
Sex	Percent Uninsured
Race	Percent College Graduate
Witnessed Arrest	Unemployment Rate
ROSC Prior to ED Arrival	Percent of Families in Poverty
Bystander AED	Percent 65 Years or Older
Bystander CPR	Percent 85 Years or Older
Shockable Initial Rhythm	Percent Female
Arrest at Home vs Public	Percent Black or African American
Onset Before vs After EMS Arrival	Average Household Size
Time of Day EMS Dispatch	Percent Vacant Housing Units
Time EMS Dispatch to Scene	Percent High School Graduate
Time EMS On Scene	
Time Scene Depart to ED	
EMS Transport Distance	

Model Building Process

Model 1: Unconditional

Model 2: Patient-Level Predictors ONLY

Model 3: Patient + Neighborhood Predictors

Model Fit Statistics (-2LL)	Model 1	Model 2	Model 3
Survival to ED	1778.7	1093.5*	1056.6*
Survival to Hospital Admission	504.3	339.5*	319.6
Survival to Discharge	340.9	202.8*	176.4*
Survival to Home Discharge	227.3	117.9*	89.0*

Neighborhood predictors significantly influence outcomes EXCEPT for survival to hospital admission

Conclusion

- Large sub-county geographic disparities exist in OHCA outcomes
- We can help explain this variation using prehospital and neighborhood socioeconomic predictors
- This research represents a replicable schematic for assessing geographic variations in survival outcomes

Future Research

- Test the model's predictive ability using current data
- Apply this model to assess geographic variation in modifiable prehospital interventions (e.g. Bystander AED)
- Expand this model to other regions to compare inter-county variation and generalizability

THANK
YOU